PROJECT PROPOSAL: OMNI-MESH

A Decentralized, Multi-Agent Framework for Unified Traffic Optimization and Edge-Based Security

ABSTRACT

Contemporary smart city traffic systems suffer from critical vulnerabilities: they rely on centralized server architectures prone to single points of failure, utilize rigid or localized sensor data that cannot adapt to dynamic urban anomalies, and operate in isolation from public safety infrastructure. This project proposes Omni-Mesh, a novel, two-tiered Multi-Agent System (MAS) that decentralizes urban traffic management. At the edge layer, individual intersections operate as autonomous nodes utilizing localized computer vision for dynamic queue assessment and peer-to-peer messaging for localized "Green Wave" emergency routing. At the cognitive layer, a global orchestrator oversees the mesh, resolving macro-level anomalies. Furthermore, this dual-layer architecture integrates a continuous Automated License Plate Recognition (ANPR) security mesh, allowing the system to seamlessly transition from traffic optimization to dynamic threat containment. By bridging decentralized physical execution with global cognitive oversight, this research aims to demonstrate a highly fault-tolerant, dual-objective urban infrastructure model.

1. CORE PROBLEM STATEMENT

Current traffic infrastructure research faces a dichotomy:

1. **Centralized Systems (Cloud-based AI):** Offer high-level global optimization but suffer from high latency, massive bandwidth costs, and catastrophic gridlock if the central server fails.
2. **Decentralized Systems (Actuated signals/Local RL):** Are robust and fast but lack macro-awareness, often pushing traffic bottlenecks to adjacent unmonitored zones or failing to coordinate complex emergency routing.

The Gap: There is a lack of hybrid architectures that can handle sub-second physical execution autonomously at the edge while simultaneously processing macro-level, city-wide anomalies (like security threats or extreme weather events).

2. PROPOSED SYSTEM ARCHITECTURE

The Omni-Mesh architecture solves this using a Neuro-Symbolic, multi-agent approach divided into two distinct tiers:

Tier 1: The Decentralized Edge (Execution & Perception)

- **Hardware:** Low-cost edge microcontrollers (e.g., Raspberry Pi 4B) deployed at individual intersections.
- **Vision-Driven State Space:** Instead of binary induction loops, the edge nodes run lightweight computer vision models (e.g., YOLOv8) to calculate dynamic queue density, vehicle classification, and wait times in real-time.
- **Peer-to-Peer (P2P) Communication:** Nodes communicate via lightweight publish-subscribe protocols (MQTT) to share state data with immediate geographical neighbors.

- **Autonomous Emergency Routing:** When an emergency vehicle is detected visually, the local node negotiates directly with downstream nodes to initiate a rolling "Green Wave," dispersing cross-traffic incrementally without requiring central server authorization.

Tier 2: The Zone Orchestrator (Cognition & Security)

- **The Global Agent:** A centralized reasoning agent that does not micromanage light timers, but monitors the grid for macro-events.
- **The ANPR Security Mesh:** The edge nodes simultaneously run Optical Character Recognition (OCR) against a localized watchlist. When a flagged vehicle is identified, the Orchestrator takes control, utilizing the MAS to predict the vehicle's trajectory and actively manipulate traffic signals ahead of the suspect to safely force a containment scenario.

3. NOVELTY AND RESEARCH CONTRIBUTIONS

If successfully implemented, this project will contribute to the field of Multi-Agent Systems by proving:

1. **Zero Single Point of Failure (ZSPF):** Demonstrating that if the Tier 2 Orchestrator goes offline, the Tier 1 nodes gracefully degrade into an autonomous P2P network, preventing system-wide traffic collapse.
2. **Dual-Objective Agent Negotiation:** Proving that an edge agent can successfully balance civilian traffic throughput (using Reinforcement Learning) with high-priority security interventions (Rule-based heuristics) simultaneously.
3. **Hardware-in-the-Loop Viability:** Validating that this complex MAS can run on resource-constrained, standard IoT hardware using Python-based ML libraries (TensorFlow/Keras) rather than requiring prohibitively expensive smart-city supercomputers.

4. PROPOSED DEVELOPMENT & EVALUATION STACK

Component	Tool / Description
Environment Simulation	SUMO (Simulation of Urban MObility) to simulate the grid, vehicle physics, and standard traffic baseline metrics.
Agent Logic & ML	Python (TensorFlow/Keras for the RL agents and vision models).
Messaging Protocol	MQTT for simulating the decentralized inter-agent communication layer.